Probability and Statistics 1B

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Contents

1 Probability Models and Random Variables

Theorem. Let X be a continuous random variable with pdf $f_X(x)$ and $\mathbb{P}(a < X < b) = 1$. Suppose $g : (a, b) \to \mathbb{R}$ is continuous, strictly increasing and differentiable.

Then $Y = g(X)$ is a continuous random variable with PDF

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y))\frac{dx}{dy} & g(a) < y < g(b) \\ 0 & y \leq g(a) \text{ or } y \geq g(b) \end{cases}.$$

2 Important Families of Continuous Random Variables

2.1 Uniform Distribution

2.2 The Normal Distribution

Definition. The random variable X has the **normal distribution** with mean μ and variance σ^2, denoted $X \sim N(\mu, \sigma^2)$ if it has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}.$$

Definition. The random variable Z is said to be **standard normal** if it follows a $N(0, 1)$ distribution.

The pdf of the standard normal is denoted by $\phi(z)$, and the cdf denoted by $\Phi(z)$.

Theorem.

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} = \sqrt{(2\pi)}.$$

Then this implies that

$$\int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{\sigma^2}} dx = \int_{-\infty}^{\infty} \sqrt{2\sigma^2} e^{-\frac{u^2}{2}} du = \sqrt{2\pi\sigma^2}.$$

Then, adding the reciprocal of this to the normal pdf guarantees it will always integrate to 1.

The standard normal, $Z \sim N(0, 1)$, has pdf ϕ and cdf Φ.

Lemma.

- $\phi(-z) = \phi(z)$
- $\Phi(-z) = 1 - \Phi(z)$.

Proposition. Suppose $Z \sim N(0,1)$ and $Y = \mu + \sigma Z$. Then $Y \sim N(\mu, \sigma^2)$.

Theorem. Let $0 < p < 1$ and $S_n \sim Binom(n,p)$. Then for any $a \in \mathbb{R}$

$$\mathbb{P}\left(\frac{S_n - np}{\sqrt{np(1-p)}} \leq a\right) \to \Phi(a) \text{ as } n \to \infty.$$

2.3 Exponential Distribution

Definition. The random variable X has an exponential distribtion with rate parameter $\lambda > 0$, denoted $\text{Exp}(\lambda)$, if it has pdf

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

and cdf

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

It has expectation

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

and variance

$$Var[x] = \frac{1}{\lambda^2}.$$

Proposition. Let X be a random variable with $X \sim Exp(\lambda)$. Then X is memoryless, meaning

$$\mathbb{P}(X > t + s | X > t) = \mathbb{P}(X > s).$$

Definition. The hazard rate at time t of survial time distribution is

$$h(t) = \lim_{\delta t \downarrow 0} \frac{\mathbb{P}(X \in (t, t+\delta t) | X > t)}{\delta t}.$$

If X is continuous, then

$$\mathbb{P}(X \in (t, t+\delta t)) \approx f_x(t).$$

Thus,

$$h(t) = \frac{f_X(t)}{1 - F_X(t)}.$$

For an exponential distribution, $h(t) = \lambda$. Indeed, the exponential distribution is derived from the properties of constant hazard rate and memorylessness.

Definition. The random variable X has a Weibull distribution with parameters λ and β, denoted $Weib(\lambda, \beta)$, if it has pdf

$$f_X(x) = \begin{cases} \lambda\beta x^{\beta-1}e^{-\lambda x^{\beta}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The value of β shapes the hazard rate function $h(t)$. For $\beta = 1, h(t)$ is constant, and $Weib(\lambda, 1) = Exp(\lambda)$. For $\beta > 1$, $h(t)$ increase with t, old items are more likely to fail than new ones, for $\beta < 1, h(t)$ decreases with t, old items are less likely to fail than new ones.

2.4 The Gamma Distribution

Definition. The Gamma function is defined for $t > 0$ as

$$\Gamma(t) = \int_0^{\infty} x^{t-1}e^{-x}dx.$$

Lemma.

$$\Gamma(t+1) = \int_0^{\infty} x^t e{-}x dx$$
$$= t\Gamma(t).$$
$$= t(t-1)\Gamma(t-1). \quad \text{and so on}$$

We can calculate that $\Gamma(1) = 1$, and hence we can show that for positive integers

$$\Gamma(t) = (t-1)!.$$

Definition. The random variable X has a Gamma distribution with parameters λ and k $(\lambda > 0, k > 0)$, denoted $Gamma(\lambda, k)$, if it has PDF

$$f_X(x) = \begin{cases} \frac{1}{\Gamma(k)}\lambda^k x^{k-1}e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

The Gamma Distribution has $\mathbb{E}[X] = \frac{k}{\lambda}$ and $Var[X] = \frac{k}{\lambda^2}$
The $\Gamma(\lambda, 1)$ distribution is equal to the $Exp(\lambda)$ distribution.

Proposition. If $X \sim Gamma(\lambda, k)$ and $Y = cX$, then

$$Y \sim \Gamma(\frac{\lambda}{c}, k).$$

Proposition. If $X_1, ..., X_k$ are independent variables where $X_i \sim Exp(\lambda)$, then

$$\sum_{i=1}^{k} X_i \sim \Gamma(\lambda, k).$$

Hence for integers k_1 and k_2, if $Y_1 \sim \Gamma(\lambda, k_1)$ and $Y_2 \sim \Gamma(\lambda, k_2)$ then

$$Y_1 + Y_2 \sim \Gamma(\lambda, k_1 + k_2).$$

Proposition.

- If $X \sim N(0,1)$, then $X^2 \sim \Gamma(\frac{1}{2}, \frac{1}{2})$.
- If $X_1, ..., X_k$ are independent $N(0,1)$ random variables, then

$$\sum_{i=1}^{k} X_i^2 \sim \Gamma(\frac{1}{2}, \frac{k}{2}).$$

The second distribution is also known as the χ^2 distribution on k degrees of freedom.

3 Generating a Random Sample from a given Distribution

3.1 Methods for Generating $Unif(0,1)$ Random Variables

If, for some large m, we can randomly arrange the numbers in the set $\{1, ..., m-1\}$ into a sequence $X_1, X_2, ...$, then we can treat $\frac{X_1}{m}, \frac{X_2}{m}, ...$ as samples from an independent $Unif(0,1)$ distribution.

Example. A way to do this is the following algorithm:

Set

$$m = 2^N \quad \text{for a large value of } N$$

Choose

$$c \quad \text{a prime number}$$

$$a \quad \text{such that } a-1 \text{ is divisible by c.}$$

Set X_1 to be any value in $\{0, ..., m-1\}$
Then proceed iteratively setting

$$X_{n+1} = (aX_n + c) \pmod{n}.$$

3.2 The Inverse CDF Method for Simulation

To turn a random sample from a $Unif(0,1)$ RV into a sample from a RV with CDF $g(x)$.

Consider the case where $g(x)$ is continuous and strictly increasing $g : [a, b] \to [0, 1]$ with $g(a) = 0$ and $g(b) = 1$.

Then the function $g^{-1} : [0, 1] \to [a, b]$ is well defined.

Lemma. Suppose the continuous random variable X has CDF $G(x)$, and $Y = G(X)$.
Then $Y \sim Unif(0, 1)$.

Theorem. Let $G(x)$ is a continuous, strictly increasing function from $[a, b] \to [0, 1]$ with $G(a) = 0$ and $G(b) = 1$. The cases $a = -\infty$ and $b = \infty$ are also allowed.
Let $U \sim Unif(0, 1)$ and set $X = G^{-1}(U)$.
Then X is a random variable with CDF $G(x)$.

4 Modelling Data and Parameter Estimation

4.1 Estimation by the Method of Moments

We start with the simple case of modelling the distribution of a single random variable.

Here, we have to choose a suitable distribution and estimate parameters of this distribution.

The method of moments tells us to fit the properties of chosen distribution to the properties of our sample, for example fitting $\mathbb{E}[X]$ and $Var[X]$.

Example. The distance between two fish in a tank is believed to follow a $Beta(a, b)$ distribution. To estimate a and b, we need two pieces of information, for example, $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
With the observed values $x_1, ..., x_n$, we can calculate $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$, using these to find a and b.

Definition. For $k \geq 0$, the kth moment of a random variable X is

$$\mu_k = \mathbb{E}[X^k] \quad \text{(if this exists)}.$$

We assume $X_1, ..., X_n$ are i.i.d. (independent, identically distributed) and follow a distribution with unknown parameters $\theta_1, ..., \theta_n$. We can then estimate $\mu_1, ..., \mu_k$ from the observed data and use these to calculate $\theta_1, ..., \theta_n$.

Definition. Method of Moments Estimates of $\theta_1, ..., \theta_p$ are the solutions to

$$\frac{1}{n}\sum_{i=1}^{n} x_i^k = \mu_k = \mu_k(\theta_1, ..., \theta_p) \text{ for k } = 1, ..., p$$

where $x_1, ..., x_n$ denote the observed values of $X_1, ..., X_n$

5 Estimates and Estimators

Definition. An **estimate** is a real number computed from the data. An **estimate** of the paramter θ based on observations $x_1, ..., x_n$ can be written as

$$\hat{\theta} = h(x_1, ..., x_n)$$

for an appropriate function h.

Definition. An **estimator** is a random variable, and is a function of the random variables $X_1, ..., X_n$ that comprise the data. If the **estimate** of θ based on observations $x_1, ..., x_n$ is $\hat{\theta} = h(x_1, ..., x_n)$, the **estimator** T is the random variable

$$T = h(X_1, ..., X_n).$$

5.1 Maximum Likelihook Estimation

Suppose the random variables $X_1, ..., X_n$ follow a certain type of distribution but the parameters of said distribution are unknown.

Definition. Let $X_1, ..., X_n$ be i.i.d discrete random variables with PMF

$$P(X_i = x) = p_X(\theta, x_i) \quad x_i \in \Omega \quad \text{for } i = 1, ..., n$$

where θ denotes the parameter or a vector of parameters.

SSyppose values $X_i = x_i, i = 1, ..., n$ are observed, and let $bfx = (x_1, ..., x_n)$.

The likelihood function for these discrete random variables is

$$L(\theta, x) = \prod_{i=1}^{n} p_X(\theta, x_i).$$

Definition. Let $X_1, ..., X_n$ be i.i.d continuous random variables with PDF $f_X(\theta, x_i)$ where θ denotes the parameter or a vector of parameters.

Syppose values $X_i = x_i, i = 1, ..., n$ are observed, and let $bfx = (x_1, ..., x_n)$.

The likelihood function for these discrete random variables is

$$L(\theta, x) = \prod_{i=1}^{n} p_X(\theta, x_i).$$

Definition. In both the continuous and discrete cases, the log-likelihood function is

$$\mathcal{L}(\theta, x) = log\{L(\theta, x)\}.$$

Definition. The **Maximum Likelihood Estimate** (MLE) of a parameter of vector of parameters, θ, is the value of θ that maximises the liklihood function $L(\theta, \mathbf{x})$ for the observed data.

5.2 Assesment of Goodness of Fit

Definition. Suppose we observe i.i.d. RVs $X_1, ..., X_n$. Denote the ordered values of the observed data $x_1, ..., x_n$ by

$$x_{(1)} \leq ... \leq x(n).$$

Let F_X be the CDF of a distribution proposed for the RVs X_i. Then, a **Quantile-Quantile plot** is a graph of

$$x_{(i)} \text{ against } F_X^{-1}\left(\frac{i}{n+1}\right).$$

If the RVs $X_1, ..., X_n$ do indeed come from the distribution with CDF $F_X(x)$, points in the Q-Q plot should lie close to a straight line through the origin with gradient 1.

6 Joint Distributions

6.1 Joint PMFs and Joint PDFs

Definition. We say random variables X and Y are jointly continuous if there eists a function $f_{X,Y} : \mathbb{R}^2 \to [0, \infty]$ such that for any region $A \subseteq \mathbb{R}^2$

$$\mathbb{P}((X, Y) \in A) = \int \int_A f_{X,Y}(x, y)\ dy\ dx.$$

The function $f_{X,Y}$ is called the joint PDF of X and Y.

Definition. The joint CDF of random variables X and Y is the function $F_{X,Y} : \mathbb{R}^2 \to [0, 1]$ defined by

$$F_{X,Y} = \mathbb{P}(X \leq x, Y \leq y).$$

If X and Y are jointly continuous random variables with joint PDF $f_{X,Y}$ then

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\ dv\ du.$$

6.2 Marginal PDFs

Proposition. If X and Y have joint PDF $f_{X,Y}$, then X and Y have PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy.$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx.$$

Definition. Let X and Y be continuous random variables with joint PDF $f_{X,Y}$ and marginal PDFs f_X and f_Y. For a value x with $f_X(x) > 0$, the **conditional pdf** of Y given $X = x$, written $f_{Y|X}(y|X = x)$ is

$$f_{Y|X}(y|x) = f_{Y|X}(y|X = x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \quad \text{for } y \in \mathbb{R}$$

7 Independence

Definition. The random variables X and Y are **independent** if

$$\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq Y)F_{X,Y}(x,y) \quad = F_X(x)F_Y(y).$$

Theorem. Suppose the random variables X and Y are jointly continuous with joint PDF $f_{X,Y}(x,y)$ and marginal PDFs $f_X(x)$ and $f_Y(y)$. If

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

then X and Y are Independent.

Definition. We say $X_1, ..., X_n$ are jointly continuous random variables if there exists a function $f_{X_1,...,X_n}(x_1, ..., x_n) : \mathbb{R}^n \to [0, \infty)$ such that for any region $A \subseteq \mathbb{R}^n$

$$\mathbb{P}((X_1, ..., X_n) \in A) = \int \int_A f_{X_1,...,X_n}(x_1, ..., x_n)\ dx_1\ dx_n.$$

Then $f_{X_1,...,X_n}$ is called the joint PDF of $X_1, ..., X_n$

Remark. If $X_1, ..., X_n$ are independent then

$$f_{X_1,...,X_n}(x_1, ..., x_n) = \prod_{i=1}^{n} f_{X_i}(x_i).$$

7.1 Expectation with Two Random Variables

Theorem. Suppose X and Y are continuous random variables with joint PDF $f_{X,Y}$ and $h : \mathbb{R}^2 \to \mathbb{R}$ is a continuous function. Then

$$\mathbb{E}[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) f_{X,Y}(x,y)\ dy\ dx.$$

as long as $\mathbb{E}[|h(X,Y)|] < \infty$.

Proposition. If X and Y are jointly continuous Random Variables, $a \in \mathbb{R}$ and $b \in \mathbb{R}$, then

$$\mathbb{E}[aX + bY] = a\mathbb{E}(X) + b\mathbb{E}(Y).$$

Proposition. Suppose X and Y are independent, jointly continuous random variables and g and h are continuous functions defined $\mathbb{R} \to \mathbb{R}$, then if the relevant integrals exist

- $\mathbb{E}[XY] = \mathbb{E}(X)\mathbb{E}(Y)$
- $\mathbb{E}[g(X)g(Y)] = \mathbb{E}(g(X))\mathbb{E}(h(Y))$

7.2 Covariance

Definition. The **covariance** of X and Y is

$$Cov(X,Y) = \mathbb{E}([X - \mathbb{E}[X]][Y - \mathbb{E}[Y]]).$$

Remark. Note that

$$Cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Definition. The **correlation** between X and Y is

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}.$$

Proposition.

- $Cov(X,X) = Var(X)$
- $Cov(X,Y) = Cov(Y,X)$
- $Cov(aX + bY, Z) = aCov(X,Z) + bCov(Y,Z$
- $Var(aX + bY) = a^2Var(X) + 2abCov(X,Y) + b^2Var(Y)$
- $-1 \leq Corr(X,Y) \leq 1$

- $Corr(X,Y) = \pm 1 \Leftrightarrow Y = aX + b$ with a of the same sign
- If X and Y are independent then $Cov(X,Y) = 0$, but the converse is false

8 Sums of Random Variables

Proposition.

$$\mathbb{E}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathbb{E}(X_i).$$

Proposition.

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + 2\sum \sum_{1\leq i<j\leq n} Cov(X_i, X_j).$$

Proposition. Suppose $X_1, ..., X_n$ are independent random variables. Then

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i).$$

Proposition. Suppose $X_1, ..., X_i$ are independent, identically distributed random variables. Then

$$Var(\bar{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{Var(X_i)}{n}.$$

Theorem. Let X, Y be random variables, and let $W = X + Y$. Then W has pdf

$$f_W(w) = \int_{-\infty}^{\infty} f_{X,Y}(x, w - x)dx.$$

Proposition. If X and Y are independent random variables, then their sum $W = x + Y$ has PDF

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x) f_Y(w - x)dx.$$

Proposition. If X and Y are continuous random variables taking positive values, if $X, Y > 0$ for all X then w has pdf

$$f_W(w) = \int_{0}^{w} f_{X,Y}(x, w - x).$$

8.1 Sums of Normal Distributions

Proposition. If X and Y are independent and

$$X \sim N(\mu_X, \sigma_X^2) \text{ and } Y \sim N(\mu_Y, \sigma_Y^2).$$

Then

$$W = X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2.$$

Proposition. Suppose $X_i, ..., X_n$ are i.i.d. with $X_i \sim N(\mu, \sigma^2)$. Then

$$\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2).$$

8.2 Central Limit Theorem

Theorem. Suppose $X_1, X_2, ...$ are i.i.d. with finite expectation and variance, with $\mathbb{E}(X_i) = \mu$ and $Var(X_i) = \sigma^2 > 0$. Let $S_n = X_1 + ... + X_n$. Then, for any number a

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} < a\right) \to \Phi(a) \text{ as } n \to \infty.$$

9 Evaluation Estimators

Definition. The **sampling distribution** of an estimator $T(X_1, ..., X_n)$ is the distribution of the random variable T, which follows from the definition of T and the joint distribution of $X_1, ..., X_n$.

Definition. The **bias** of an estimator $T(X_1, ..., X_n)$ of θ is

$$\text{Bias}(T) = \mathbb{E}[T] - \theta.$$

An estimator $T(X_1, ..., X_n)$ of θ is **unbiased** if

$$\mathbb{E}[\theta] = \theta \quad \text{ for all } \theta.$$

Definition. The **precision** of an estimator $T(X_1, ..., X_n)$ of θ is

$$\text{Precision}(T) = \frac{1}{\text{Var}(T)}.$$

Definition. The **mean squared error** of an estimator $T(X_1, ..., X_n)$ of θ is

$$\text{MSE}(T) = \mathbb{E}[(T - \theta)^2].$$

Remark. The mean sqaure error can be calculated by

$$\mathrm{MSE}(T) = \mathrm{Var}(T) + (\mathrm{Bias}(T))^2.$$

There is a simple way to modify the maximum liklihood estimate to make it unbiased; define a new estimator T', where $\mathbb{E}[T'] = \theta$.